

Sankhojjal Chatterjee

+91-7003889627 | sankhojjal@gmail.com | linkedin.com/in/sankhojjalchatterjee | github.com/sankhojjalc | sankhojjal-dev.web.app

SUMMARY

Senior Software Engineer with **9+ years** of experience designing and scaling enterprise systems across **banking and e-commerce**, with a focus on **Applied AI, Generative AI, and Agentic AI**. Experienced in building **RAG systems, LLM-powered workflows, AI agents, evaluation pipelines, Python/FastAPI services, secure tool integrations, and React/TypeScript applications**. Strong background in enterprise integrations, production reliability, observability, security, and translating complex business workflows into scalable technical solutions.

TECHNICAL SKILLS

Applied AI & Agents: LLMs, RAG, Agentic AI, LangGraph, Tool / Function Calling, Human-in-the-Loop, Prompt Engineering, Context Engineering, Structured Outputs, AI Guardrails

Retrieval & Evaluation: Embeddings, Vector Search, BM25, Hybrid Search, Metadata Filtering, Reranking, Chunking, RAG Evaluation, LLM-as-a-Judge, AI Observability

Backend & Data: Python, FastAPI, Pydantic, Node.js, REST APIs, GraphQL, PostgreSQL, Redis, Kafka

Frontend: React.js, TypeScript, JavaScript, Redux Toolkit, Module Federation, Webpack, SSR, Progressive Web Apps

Cloud & DevOps: AWS, Docker, Kubernetes, CI/CD, GitHub Actions

Security & Reliability: OAuth 2.0, Keycloak, RBAC, Prompt Injection Protection, Model/Tool Guardrails, Sentry

EXPERIENCE

SBER Bank

Senior Software Engineer

January 2025 – Present

Bengaluru, India

- Designed an **AI-powered Cross-Border Payments & Compliance Copilot** combining RAG, tool calling, and agentic workflows for transaction investigation, regulatory-policy discovery, document analysis, and case resolution, reducing average manual investigation time by approximately **31%** across supported payment scenarios.
- Built **Python/FastAPI and LangGraph-based AI services** integrating LLMs with transaction, customer/KYC, document, and case-management APIs through controlled tool calling and stateful workflows, achieving **96% tool-call accuracy** while maintaining approximately **4.3s P95 latency** for standard investigation flows.
- Implemented **hybrid retrieval using dense embeddings, BM25, metadata filtering, and reranking** across regulatory policies, banking procedures, and operational knowledge, achieving **87% Recall@5** and **93% groundedness** on representative evaluation datasets.
- Designed production controls including **human-in-the-loop approvals, Keycloak/RBAC authorization, Pydantic validation, tool-level permissions, prompt-injection safeguards, and audit logging**, ensuring sensitive banking actions remain outside autonomous LLM execution.
- Established an **AI evaluation and observability framework** measuring retrieval quality, groundedness, tool-call accuracy, workflow completion, latency, token consumption, and failure patterns to support controlled releases.
- Partnered with **operations, compliance, security, product, and platform engineering teams** to translate cross-border payment workflows into reusable AI architectures, evaluation criteria, and production-ready implementation patterns.

Publicis Sapient

Senior Associate Engineer

August 2021 – December 2024

London, UK / Bengaluru, India

- Developed high-scale full-stack applications using **React, TypeScript, Python/FastAPI, REST APIs, and PostgreSQL** for a UK e-commerce platform serving **20M+ monthly active users**.
- Designed end-to-end features across **React/TypeScript frontends and Python/FastAPI backend services**, defining REST contracts, request validation, pagination/filtering, error handling, and PostgreSQL persistence for **12+ high-volume customer workflows**.
- Contributed to an internal **RAG-based knowledge assistant**, implementing document ingestion, chunking, embeddings, semantic retrieval, prompt engineering, and LLM integration, reducing internal information-discovery time by approximately **35%**.
- Led **SSR/PWA architecture, Webpack optimization, caching, and code splitting**, improving page-load performance by **37%** and contributing to a **42% increase in traffic**.

Josh Software Pvt. Ltd.

Senior Engineer

August 2020 – August 2021

Pune, India

- Built full-stack solutions using **React.js, Node.js, REST APIs, and AWS** for a leading food-delivery platform processing **2M+ monthly transactions** while maintaining **99.9% uptime** during peak order periods.
- Implemented **Kafka-based event streaming** for real-time order updates and migrated backend services to AWS using Terraform, enabling zero-downtime deployments and scaling to handle **300% traffic spikes** during promotional events.
- Built real-time driver tracking using **WebSockets and Pusher**, processing **50,000+ location updates per minute** to support continuously updated delivery workflows.

Mindtree Ltd.

Software Engineer

January 2017 – August 2020

Bengaluru, India

- Built multi-tenant enterprise applications using **React.js, JavaScript, GraphQL, Apollo Client, and reusable component-driven architecture**.
- Implemented **OAuth 2.0 and role-based access control** for secure authentication, protected routing, and permission-driven application workflows.
- Developed responsive, reusable Web Components aligned with **WCAG 2.1 accessibility standards**, improving consistency across application modules and supported screen sizes.
- Optimized frontend performance using asynchronous loading, CDN caching, and GraphQL query tuning while monitoring API efficiency through **Apollo Studio**.

EDUCATION

Techno India College of Technology

Bachelor of Technology

August 2012 – August 2016

West Bengal University of Technology